

Class19: Pertussis Mini project

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Background

Pertussis is a bacterial lung infection also known as Whooping cough. Let's begin by examining CDC reported case numbers in the US.

```
cdc <- data.frame(  
  year = c(1922L,1923L,1924L,1925L,  
            1926L,1927L,1928L,1929L,1930L,1931L,  
            1932L,1933L,1934L,1935L,1936L,  
            1937L,1938L,1939L,1940L,1941L,1942L,  
            1943L,1944L,1945L,1946L,1947L,  
            1948L,1949L,1950L,1951L,1952L,  
            1953L,1954L,1955L,1956L,1957L,1958L,  
            1959L,1960L,1961L,1962L,1963L,  
            1964L,1965L,1966L,1967L,1968L,1969L,  
            1970L,1971L,1972L,1973L,1974L,  
            1975L,1976L,1977L,1978L,1979L,1980L,  
            1981L,1982L,1983L,1984L,1985L,  
            1986L,1987L,1988L,1989L,1990L,  
            1991L,1992L,1993L,1994L,1995L,1996L,  
            1997L,1998L,1999L,2000L,2001L,  
            2002L,2003L,2004L,2005L,2006L,2007L,  
            2008L,2009L,2010L,2011L,2012L,  
            2013L,2014L,2015L,2016L,2017L,2018L,  
            2019L,2020L,2021L,2022L,2023L,2024L),  
  cases = c(107473,164191,165418,152003,  
            202210,181411,161799,197371,  
            166914,172559,215343,179135,265269,  
            180518,147237,214652,227319,103188,  
            183866,222202,191383,191890,109873,  
            133792,109860,156517,74715,69479,
```

```

120718,68687,45030,37129,60886,
62786,31732,28295,32148,40005,
14809,11468,17749,17135,13005,6799,
7717,9718,4810,3285,4249,3036,
3287,1759,2402,1738,1010,2177,2063,
1623,1730,1248,1895,2463,2276,
3589,4195,2823,3450,4157,4570,
2719,4083,6586,4617,5137,7796,6564,
7405,7298,7867,7580,9771,11647,
25827,25616,15632,10454,13278,
16858,27550,18719,48277,28639,32971,
20762,17972,18975,15609,18617,
6124,2116,3044,7063,22538)
)

```

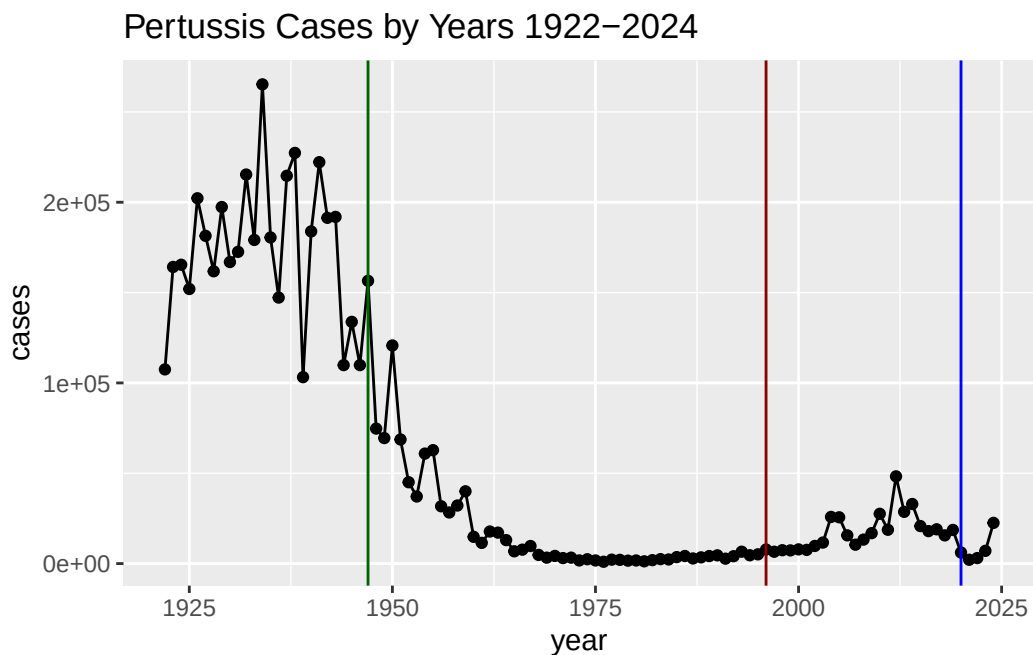
Plot of cases per year for Pertussis in the US

```

library(ggplot2)

ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() + geom_vline(xintercept = 1947, col="darkgreen") + geom_vline(xintercept = 1999, col="darkred") + geom_vline(xintercept = 2020, col="darkblue")

```



The full introduction of mandatory wP(whole-cell) Pertussis immunization in the mid 1940s leads to a dramatic reduction in case numbers (from over 200,000 to 100s)

The switch to the aP (newer acellular formalization) despite high coverage, the shorter and narrower protection from aP coincided with a gradual resurgence and periodic outbreaks in the 2000s–2010s, especially among adolescents and adults.

The 2020 lock-downs and social distancing sharply suppressed transmission, producing an unprecedented drop in reported pertussis; as measures relaxed and susceptibility accrued, incidence rebounded from 2022 through 2024.”

The CMI-PB project

The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination. Website: <https://www.cmi-pb.org>

They make their data available via JSON format API endpoints - basically the database tables in a key:value types format like “infancy_vac”:“wP”. To read this we can use the `read_json()` function from the **jsonlite** package. install with `install.packages("jsonlite")`

```
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. how many “subjects/individuals are in this datasets?”

```
nrow(subject)
```

```
[1] 172
```

Q. How many wP and aP subjects are there?

```
table(subject$infancy_vac)
```

```
aP wP  
87 85
```

Q. What is the breakdown by “biological sex” and “race”?

```
table(subject$biological_sex)
```

```
Female    Male  
    112     60
```

```
table(subject$race)
```

```
        American Indian/Alaska Native  
                1  
                Asian  
                44  
        Black or African American  
                5  
        More Than One Race  
                19  
Native Hawaiian or Other Pacific Islander  
                2  
        Unknown or Not Reported  
                21  
                White  
                80
```

```
table(subject$race,subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

This breakdown is not particularly representative of the US population - this is a serious caveat for this study. However, it is still the largest sample of its type ever assembled.

```
specimen<- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer<-read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost	
1	1	1	-3	
2	2	1	1	
3	3	1	3	
4	4	1	7	
5	5	1	11	
6	6	1	32	

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

We need to “join” or link these table with the `subject` table so we can begin to analyze this data and know who a give AB sample was collected for and when.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
meta <- inner_join(subject,specimen)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White
4	1	wP	Female	Not Hispanic or Latino	White
5	1	wP	Female	Not Hispanic or Latino	White
6	1	wP	Female	Not Hispanic or Latino	White
	year_of_birth	date_of_boost	dataset	specimen_id	
1	1986-01-01	2016-09-12	2020_dataset	1	
2	1986-01-01	2016-09-12	2020_dataset	2	
3	1986-01-01	2016-09-12	2020_dataset	3	
4	1986-01-01	2016-09-12	2020_dataset	4	
5	1986-01-01	2016-09-12	2020_dataset	5	
6	1986-01-01	2016-09-12	2020_dataset	6	
	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type		
1		-3	0		Blood
2		1	1		Blood
3		3	3		Blood
4		7	7		Blood
5		11	14		Blood

6		32		30	Blood
	visit				
1	1				
2	2				
3	3				
4	4				
5	5				
6	6				

Now let's join `ab_titer` table with our `meta` table so we have all informations about a given Ab measurement.

```
ab_data <- inner_join(meta, ab_titer)
```

Joining with ``by = join_by(specimen_id)``

```
head(ab_titer)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

Q. How many Ab measurments do we have in the total

```
nrow(ab_data)
```

```
[1] 61956
```

Q. How many different isotypes (types of Ab) are in the dataset

```
unique(ab_data$isotype)
```

```
[1] "IgE" "IgG" "IgG1" "IgG2" "IgG3" "IgG4"
```

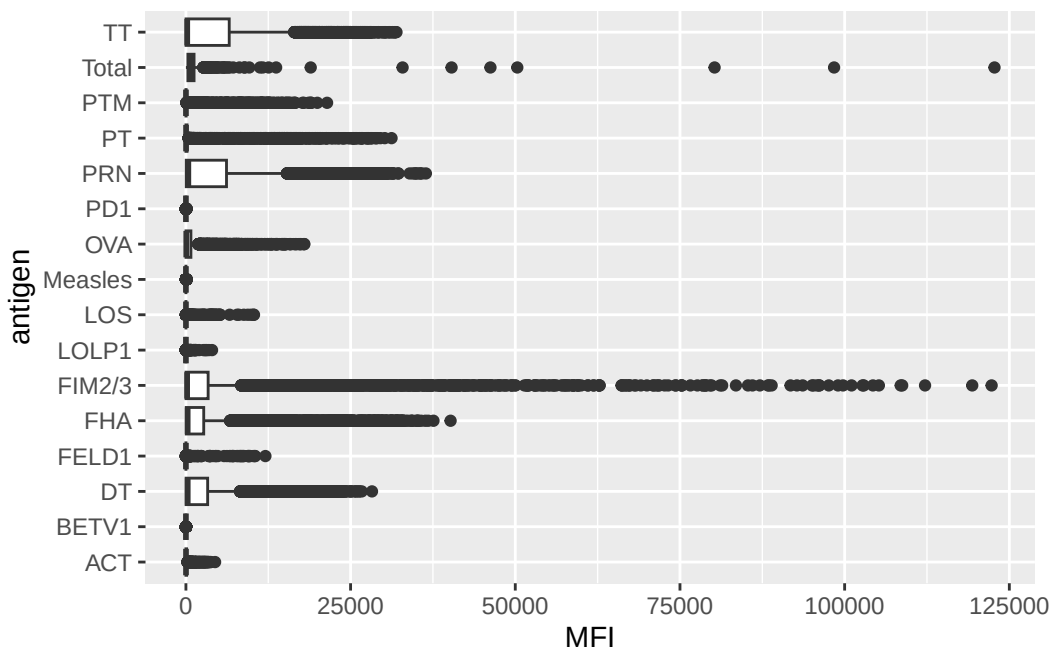
Q. How many different antigens?

```
unique(ab_data$antigen)
```

```
[1] "Total" "PT" "PRN" "FHA" "ACT" "LOS" "FELD1"  
[8] "BETV1" "LOLP1" "Measles" "PTM" "FIM2/3" "TT" "DT"  
[15] "OVA" "PD1"
```

```
library(ggplot2)  
ggplot(ab_data)+  
  aes(MFI, antigen)+  
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range (`stat\_boxplot()`).

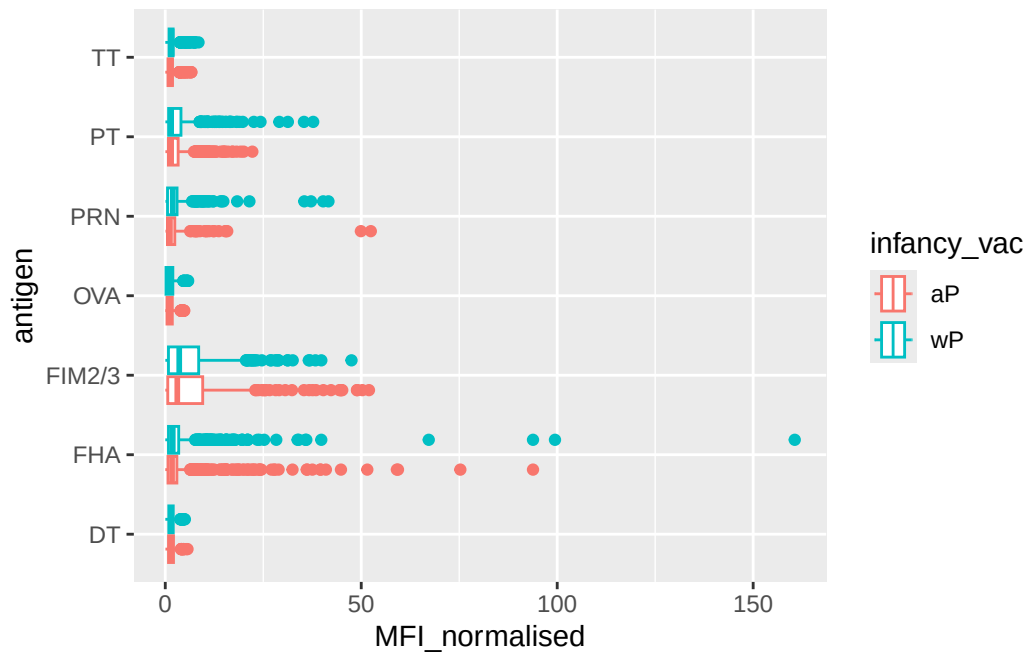


##Examine IgG Ab titer levels

IgG is crucial for long-term immunity and responding to bacterial & viral infections.

```
igg <- ab_data |>
  filter(isotype == "IgG")
```

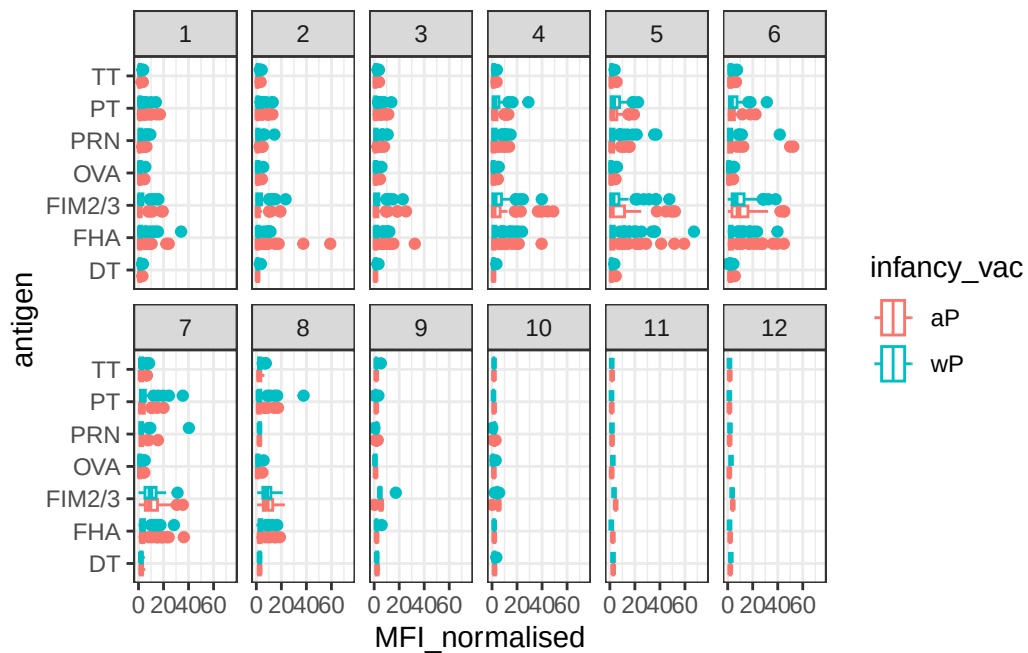
```
ggplot(igg)+
  aes(MFI_normalised, antigen,col= infancy_vac)+
  geom_boxplot()
```



We can “facet” our plot by wP vs aP

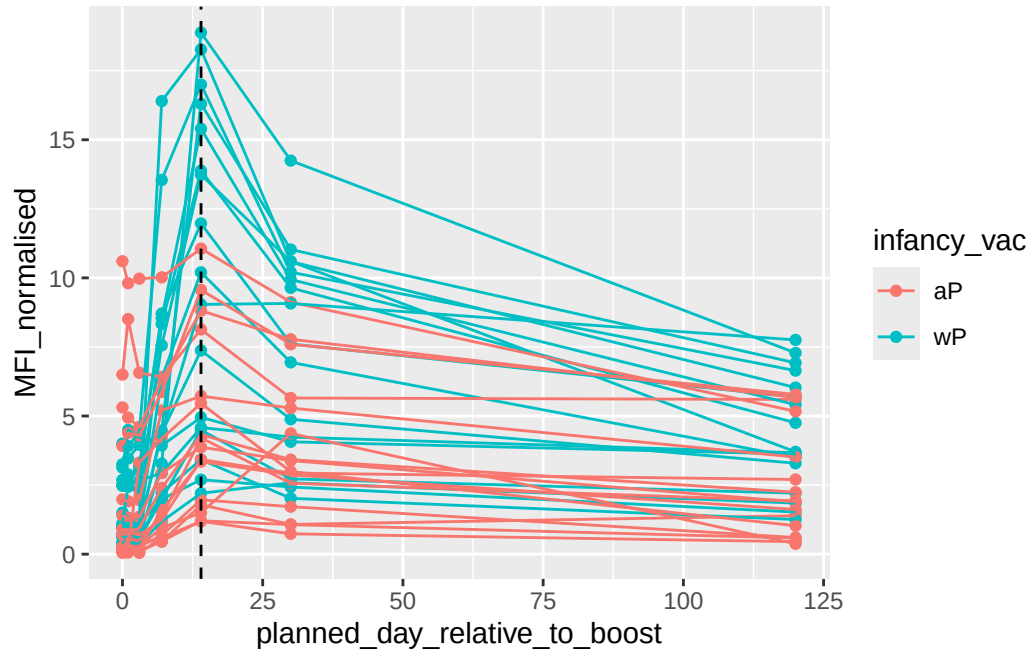
```
ggplot(igg)+
  aes(MFI_normalised, antigen,col= infancy_vac)+
  geom_boxplot()+
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



More advanced analysis digging into individual antigen responses over time:

```
filter(igg, antigen=="PT", dataset == "2021_dataset") |>
  ggplot() +
  aes(x=planned_day_relative_to_boost,
      y=MFI_normalised,
      col=infancy_vac,
      group=subject_id) +
  geom_point() +
  geom_line()+
  geom_vline(xintercept=14, linetype="dashed")
```



This plot shows the time course of Pertussis toxin (pT) antibody responses for a large set of wP (teal color), aP (red color) individuals. Levels peak at day 14 and are larger in magnitude for wP than ap individuals.

There are lots of cool things to explore in this datasets and we need coding and biology knowledge to do it effectively.